Is Token-Level the Wrong Coordinate System? Program Transformation as the Action Space for LLM-Driven Kernel Optimization

Preprint draft — see Appendix A for provenance and AI usage

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Position and analysis. Every number in Section 3 is computed from the public AdaExplore archive and is reproducible from the scripts accompanying this paper.

Abstract

Agentic reinforcement-learning systems now write GPU kernels that beat compiler autotuning. They update a policy over sampled text and attach one execution-level reward at the end of thousands of decisions. Performance, however, belongs to a handful of program design decisions — fuse two passes, stop materializing an intermediate, change the per-thread workload — each realized across many tokens and several separated code sites. We ask whether this is the wrong *coordinate system*: the action unit. We formalize a candidate action as a typed program transformation $a = (\tau, \ell, \theta)$, and separate three claims the literature conflates: description, search efficiency, learning efficiency. On public AdaExplore trajectories (2,510 successful parent→child edges; 55 annotated over 45 tasks) a frozen 12-category vocabulary covers 90.9% of edges, using 10 of its 12 categories, and under symmetric permutation calibration the semantic label predicts realised speedup beyond its own granularity ($p = 0.0005$) while coarse diff structure ($p = 0.752$) and hunk count ($p = 0.436$) do not. Three limitations bound the result: two supporting statistics are withdrawn, one as an identity and one as a sampling artefact; cross-task recurrence sits at chance; and relabelling without the category definitions leaves the headline insignificant ($p = 0.106$, $\kappa = 0.687$). K-Search, CuAsmRL, CodeAct and CUDA Agent's own skill checklist already occupy parts of the proposed interface, narrowing our contribution to measurement and protocol. Search and learning stay untested; we specify the five-condition experiment that decides them, including the branch that kills the direction.

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1. Introduction

1.1 The empirical fact that motivates the question

Modern agentic reinforcement-learning systems can now write GPU kernels that beat compiler autotuning. CUDA Agent reports 100%, 100%, and 92% faster-than-`torch.compile` rates on the three KernelBench difficulty splits, outperforming the strongest proprietary models by about 40% on the hardest split [1,2]. It reaches there by placing a language model inside a real development sandbox — compile, verify correctness, profile, iterate — and optimizing a policy over long multi-turn agent trajectories with PPO, at a context length of order 10^5 tokens [1]. The released artifacts are a 6,000-sample operation dataset, a skill specification, and the reward and warm-up training recipe; the model weights, agent trajectories, and training-time observables are not public [1,3].

What that recipe manages is the difficulty of training a long-horizon policy on rare text. The reward is a four-level quantised function $r \in \{-1, 1, 2, 3\}$ keyed to correctness and to a >5% margin over the eager and `torch.compile` baselines, chosen because "operators vary substantially in optimization difficulty, making raw speedup an unreliable proxy for code quality"; the PPO clip is asymmetric, with $\varepsilon_{\text{lower}} = 0.2$ and $\varepsilon_{\text{higher}} = 0.28$; the first run "could only train stably for 17 steps before the model's performance collapsed", and 200 stable steps came from warming up both actor and critic; the stated cause is a "severe domain distribution mismatch", because "CUDA coding data accounts for less than 0.01% of pretraining data", which yields low-probability tokens near $\pi_\theta \approx 10^{-9}$ whose importance ratios destabilise under train/inference precision differences [1, Sec. 5.1-5.2].

Each of those devices answers a symptom of the training substrate. None of them is derived from a property of a GPU: memory hierarchies, warp semantics, coalescing and bank conflicts appear nowhere in their justification. They appear in the *reward*, which is observed once, at the end of a trajectory, and then spread over the tokens that produced it.

Where that spreading happens is decided by what the learner is treated as having chosen, and this system describes that unit twice, in two different ways. In its reinforcement-learning formulation the action is the token: the clipped objective defines a_t as "the action (i.e., token) taken at position t" [1, Sec. 5.2]. In its environment description the action is something coarser: a toolset of Bash, Read/Write, Edit/MultiEdit, Glob, Grep, NotebookEdit, BashOutput and KillBash, where "these tools define the action space of the agent" and MultiEdit applies "multiple atomic edits within one file" [1, App. "Details of Agent Loop"]. Two candidate units of decision therefore coexist inside one working system, and the learning machinery is attached to the finer of them.

This paper asks what changes if the unit is taken instead from the program's performance model — a typed program transformation — and, ahead of that, whether such a unit can be described at all on the evidence that is public. Section 2.1 lays out the candidate units, Section 3 measures what the published trajectories support, and Section 4 specifies the experiment that would settle the rest.

1.2 The black hole in the middle

There is a structural gap between how these systems learn and what they are being asked to produce. Performance is an *executed, structural* property. A jump from 1.0× to 1.8× may be attributable to one design decision — fuse two passes and eliminate one global-memory round trip — while the other two thousand tokens are boilerplate that would have been written nearly identically anyway. The training interface, though, is a per-token policy update. Nothing in the pipeline tells the learner "the decision at step 1,537 is the one that paid". Value functions and GAE estimate credit *statistically* over token sequences.

This gap has its own research literature. A recent survey synthesises a 69-paper corpus of credit-assignment methods for LLM reinforcement learning organised precisely by *granularity* — trajectory, segment, step, token — and reports that sparse outcome rewards "say little about which token, reasoning step, tool call, memory operation, or agent caused an outcome" [35]. Current practice therefore already intervenes below the trajectory: process rewards, span-level advantage reweighting, turn-level verifiable supervision, counterfactual token credit. Almost all of that work changes the **credit target** while leaving the **action unit** at the token, so the learner still treats a single program decision as hundreds of separate actions whose joint realization it must discover statistically. Reweighting the token-level credit and re-typing the action unit are orthogonal remedies, and the second is the one that would remove the need for the first.

The standard response is better credit assignment: find the important spans, reweight them, sharpen the critic. Spans are the wrong place to look. A single optimization decision can be spread over a kernel body, an index computation, a shared-memory declaration, and host-side launch parameters; conversely, the same decision can be realized by completely different code. A locator over contiguous regions keeps missing things because it is trying to recover a *program decision unit* out of *textual locality*, and the two are not in one-to-one correspondence.

That is what pushes the question one level up.

1.3 The reframing

Why should the training system treat forty tokens as forty actions in the first place? They are one implementation of one program decision. The question therefore stops being "on which tokens should the reward land" and becomes:

> **Is the token-level action space itself the wrong coordinate system for LLM-driven kernel optimization?**

Concretely: does CUDA optimization admit a natural action space that is more stable and more transferable than tokens, and would moving the search and learning interface onto it measurably change search efficiency and learning efficiency?

We write the candidate action as

$$a = (\tau, \ell, \theta)$$

transformation *type*, *location*, *parameters*. The load-bearing structure is a many-to-many relation:

```
one semantic action   →  many tokens, many code sites
very different code   →  the same semantic action
```

Consider increasing per-thread workload from one element to two. In CUDA C++ this requires a new index expression in the kernel body (`int j = i + stride; if (j < n) ...`) *and* a host-side change of the block count from `N/T` to `N/(2T)`. The two edits are far apart in the file and neither one alone constitutes the action. That is the shape of the object we mean.

1.4 What would count, and what this paper actually claims

That transformations carry more semantics than tokens is not a new idea: Halide, TVM, and Ansor already put schedule transformations at the centre of their search spaces [8,9,10]. The claim here is specific to LLM-in-the-loop optimization:

> Moving the interface from *generate the whole program* to *select a parameterized transformation, then realize it* measurably changes search efficiency and learning efficiency.

And it has a prerequisite that must be checked before any of that is worth building:

> **Can real successful kernel-optimization trajectories be explained by a small set of recurring program transformations?**

If every successful child is a one-off rewrite that cannot be grouped into a stable vocabulary, the direction is dead and no amount of infrastructure fixes that. If, instead, edges keep landing on recurring patterns — increase work per thread, eliminate an intermediate, merge kernels, retile shared memory, change reduction structure — and the same pattern shows up with very different code, then we get the phenomenon worth studying: **high diff entropy, lower transformation entropy**. That is the phenomenon the measurement in Section 3 looks for.

This paper does three things:

1. **States the thesis precisely and separates claims that are routinely conflated** (Section 2): the *descriptive* claim that transformations are a stable description unit, the *search* claim that they change search efficiency, and the *learning* claim that they change learning efficiency. The first is necessary but in no way sufficient for the other two. It also separates the transformation cut from the neighbouring *turn-level* and *edit-level* cuts, both of which already exist in the literature [30,32].
2. **Reports a calibrated measurement study on public trajectories** (Section 3). The descriptive claim survives a symmetric permutation test; the transferability claim does not survive its own null.
3. **Specifies a falsifiable experimental framework** (Section 4): five conditions including an enumeration-plus-cost-model arm, and a reading tree under which each possible pattern of results implies a specific, pre-committed conclusion — including patterns that kill the direction outright. Since the protocol is not novel as a *system* — K-Search has already realized its structured-action condition with

positive results [29] — its value is as an instrument for *independently falsifying* that class of result rather than as a proposal to be first.

1.5 Scope

This is a position-and-analysis contribution: a thesis, preliminary observational evidence, and a protocol design. It contains **no training experiments, no new algorithm, and no search experiments**. The annotations behind Section 3 are machine-produced, and the circularity that creates is the subject of Section 7.1. Sections 3.6 and 3.7 report the negative results that bound the scope of the programme. Provenance and AI usage are recorded in Appendix A.

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2. Position: The Wrong Coordinate System

2.1 Three levels at which a training interface can be cut

Any optimization loop over programs must decide what counts as one action. Three cuts are natural.

Cut	Action unit	Credit lands on	Failure mode
Token	one sampled token	GAE/value estimates over $\sim 10^3$–10^4 steps	decision unit invisible; credit is statistical guesswork
Edit span / code block	a contiguous diff region	the span	decisions are *not* contiguous; spans overlap and split arbitrarily
Turn / tool call	one agent step in a multi-turn loop	the turn	coarser, but a turn still bundles several independent decisions (cf. multi-turn kernel RL [32])
Transformation	typed (τ, ℓ, θ)	the selected transformation	granularity is a design choice; realization may be the real bottleneck

The turn row is occupied: recent multi-turn kernel RL attributes reward at the turn level and reports that this matters [32]. The field has moved up from tokens by one level, not onto a typed program-transformation space. And the rows are not mutually exclusive even within one system: CUDA Agent's *interface* is edit- and tool-shaped, since its documented action space is a set of operations that includes MultiEdit ("multiple atomic edits within one file"), while the policy selecting them is trained at token granularity [1]. One system occupying two rows at once is the clearest available evidence that the coordinate system is a real design variable rather than a by-product of whichever framework is in use. The claim concerns the fourth row, and Section 4 is built to distinguish it from the third.

Most "semantic" code-RL work actually sits on the edit-span row, which is not the transformation row. An edit span is defined by the diff algorithm; a transformation is defined by program intent. Diff hunks merge when context lines happen to repeat and split when a single intent crosses a blank line. Building actions on them means building on the artefact of a text-comparison routine.

2.2 Formal shape

Let P be a program, ΔP a change, and

$$\varphi : \Delta P \to a$$

a mapping from realized diffs to action objects. The research question is whether a *simple* φ exists whose range is small, stable across tasks, and faithful to what actually caused the performance change.

An action is realized by a set of distributed edits, not a location:

$$a \to \{e_1, e_2, e_3, \dots\}$$

This is why "the optimization code" is the wrong target: it is a search for a contiguous region carrying a non-contiguous decision.

2.3 Granularity: the hard theoretical problem

The action space is only useful between two failure poles.

- **Too coarse** — `memory optimization`, `improve ILP`. Every CUDA kernel is affected; nothing is constrained; no realization obligation follows.
- **Too fine** — `change line 172 from i to i+stride`. This is a diff; entropy collapses back onto the text.

A good action is more abstract than a token edit yet concrete enough to *constrain implementation*. *work_per_thread* : $1 \to 2$ qualifies: definite semantics, direct hardware consequence, recurs across kernels, obliges coordinated change at several sites, and is far more specific than "improve ILP".

We therefore expect the eventual object to be a small typed DSL rather than a list of nouns:

```
IncreaseWorkPerThread(axis, k)    Tile(axis, size, mapping)      Fuse(producer,
↪ consumer)
EliminateMaterialization(site)    Vectorize(axis, width)         WarpReduce(scope)
```

but building that DSL before establishing that the phenomenon exists would produce engineering that answers no question, so this paper does not build it.

The test for whether a candidate sits at the right granularity can be stated without a vocabulary: an abstraction is worth keeping when it is *purely a contract* rather than a convenience for human readers. Sul and Ré make the same distinction from the other side when they argue that CUDA DSLs may retire as cognitive offloaders but hit "a floor where an abstraction is purely the contract, rather than an offloader, and it stops" [36]. Translated to actions: `Fuse(producer, consumer)` is a contract because an implementer cannot satisfy it without touching the sites the fusion requires, whereas `memory optimization` is only a label — nothing fails to count as it. That is also why the cross-site obligation of Section 2.2 matters so much: it is the operational content of "contract".

2.4 Why this is not just "plan first"

A prompt that says "describe your plan, then write the kernel" is not evidence for anything interesting. It adds reasoning tokens and may help for entirely generic reasons. The distinction we need is structural:

```
plan-first:      P → free-text reasoning → P'
transformation:  P_t --a_t--> P_{t+1},   a_t ∈ A, reusable, comparable, statable,
↪  recurring across tasks
```

Free-text reasoning has no learnable structure. A typed action set does: it supports $Q(P, a)$ instead of $Q(P, \text{one particular token string})$. Any experimental claim must therefore include a free-plan control, or it proves nothing. Section 4 builds that control in.

2.5 What "transferable" should mean here

Transfer here means something narrower than cross-vendor portability (NVIDIA → AMD) or cross-architecture generalization. An optimization written for one vendor's memory system is not falsified by failing on another's; demanding cross-vendor generality confuses a hardware claim with a modelling claim.

The transfer we care about is *within* one target platform, across kernels:

$code_A \neq code_B$ but $a_A = a_B =$ WarpReduce

so that experience collected on reduction A can be indexed by action rather than by surface code and reused on reduction B, turning $Q(P, \text{token string})$ into $Q(P, a)$.

Published cross-vendor results bear on this only indirectly. NKI-Agent ports the CUDA-Agent framework to AWS Trainium and reports that GRPO with a binary compilation reward *fails to improve on supervised fine-tuning* [18]. The reinforcement-learning component therefore did not carry over even where the agent interface did, which makes "the recipe transfers" an assumption to be tested rather than a premise.

The evidence for this version of transferability is not in the data used here. Section 3.5 shows that cross-task recurrence of the transformation labels sits at chance under a label-permutation null, because the sampling caps edges per task. The transferability reading of the thesis is therefore **not supported by this data** — it is a hypothesis that Section 4 would have to earn.

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3. Preliminary Evidence: Do Successful Optimizations Compress?

3.1 Data and why this corpus

We use AdaExplore's public parent→child optimization trajectories [3], collected on KernelBench [2]. This choice is forced rather than convenient. CUDA Agent, the system that motivates the critique, does not release agent trajectories, generated versions, tool calls, feedback logs, per-token critic values, raw probabilities, or advantages; requests for this in the public issue tracker have gone unanswered. Any study that needs the *training-time observables* of that system is simply not executable

by outsiders [1]. AdaExplore is a public search trajectory corpus, which is what the prerequisite question actually needs.

Archive structure per task: `step_N.py` (the `ModelNew` source), `step_N_log.json` (MCTS node metadata), `step_N_prompt.txt`, `step_N_metrics.json` (correctness and timing).

Corpus audit (before any labeling). 23,731 source files across 150 task directories: 22,007 contain Triton kernels, **0 contain CUDA C++**, and 1,724 contain neither (PyTorch baselines). The available public trajectories are *Triton* trajectories. Triton hides `threadIdx`, shared memory, and warp shuffles behind a block-level API, so two of our twelve categories (`WarpReduce`, `SharedMemoryTransform`) have no directly observable realization, and they are indeed the two that never appear as a primary label (Section 3.5). The claim we can test is therefore about *schedule-level* transformations, not thread-level CUDA ones.

3.2 Edge definition

An edge is a genuine MCTS parent→child pair taken from the explicit `parent_node_id` field — *not* from adjacent step numbers, which would mix tree children with unrelated siblings. Eligibility: both endpoints compiled and correct, and the child more than 5% faster (measured `runtime_stats.mean`). The corpus contains **2,510** eligible edges (1,906 on Level 2, 604 on Level 3) across 145 task trees.

We annotate **55** edges across 45 tasks (49 Level 2, 6 Level 3), at most three per task, selected toward the largest improvements. Sampling is deliberately enriched for multi-hunk edges (33 of 55), because "one action, several sites" is the object of the test and a uniform sample would make it hard to observe.

This is a biased sample and the bias has two consequences for how these numbers may be used. It cannot estimate the population label distribution, so the red lines of Section 3.4 are descriptive statements about these 55 edges rather than falsification tests of the programme. Worse, the per-task cap of three edges saturates the maximum attainable distinct-task count: with 55 edges over 45 tasks, essentially *any* assignment of labels to edges spreads across tasks. No cross-task-recurrence claim is made (Section 3.5).

3.3 Frozen vocabulary

Twelve categories, frozen *before* any edge diff was read. The freeze used corpus-level metadata only (directory layout, JSON fields, the language census above) and no edge content, which is what would constitute leakage. Category names and their count were fixed at that point; their *definitions* are stated as schedule-level semantics with two alternative realizations (CUDA C++ / Triton), because Section 3.1 shows the data contains no CUDA C++.

The twelve: `Fuse`, `EliminateMaterialization`, `Tile`, `IncreaseWorkPerThread`, `Vectorize`, `ChangeLaunchConfig`, `WarpReduce`, `SharedMemoryTransform`, `PrecisionChange`, `LoopTransform`, `HostDeviceOverlap`, `Other`.

Labeling is multi-label with one designated primary, an argument field θ where recoverable, a list of code regions the action lands on, and a free-text justification that

must quote concrete changed lines. `Other` requires an explicit account of why each of the eleven specific categories was rejected.

Annotation is machine-generated, not human: an automated annotator read and labelled the edges. That creates a circularity risk — the vocabulary and the annotator come from the same cognitive family, so the labelling procedure is not independent of the hypothesis it tests — which Section 7.1 states, together with the acceptance criteria that would control it.

3.4 Red lines that were specified in advance

Two diagnostic thresholds, fixed before the labels were read:

Condition	Threshold	Observed	Fired?
Any single category $>$40% of edges ⇒ vocabulary too coarse	40%	`Fuse` 30.9% (17/55)	**no**
`Other` $>$25% ⇒ the conjecture may be false	25%	9.1% (5/55)	**no**

Neither threshold fired *numerically*. Both are weaker than they look, for two reasons.

- The thresholds were designed as population-level falsification criteria but are being evaluated on the multi-hunk-enriched, per-task-capped sample of Section 3.2, which cannot estimate a population distribution. Reading "not fired" as "the vocabulary is neither too coarse nor fictional" would be an over-claim.
- The `Other` rate understates the conceptual failure. All five `Other` edges turn out to be the *same* missing action (Section 3.7a): a vocabulary that needs one new category to absorb 100% of its refusals has a structural hole, even though 9.1% of edges is comfortably below 25%. The numerical red line was passed by a statistic that the qualitative evidence contradicts.

3.5 Result: what the descriptive premise can rest on

The prerequisite question gets a qualified yes.

Quantity	Value	Status
Vocabulary reuse rate $\|V\|/\|E\|$	10 / 55 = 0.18	meaningful
Fraction of edges not requiring `Other`	90.9%	meaningful, but see 3.4
$H(\text{primary label})$	2.67 bits; 0.805 of its own maximum (10 cells)	meaningful
$H(\text{coarse structure descriptor, 24 cells})$	4.26 bits; 0.928 of maximum	meaningful comparator
$H(\text{diff fingerprint})$	5.78 bits = $\log_2 55$ exactly	**identity, not evidence**
Edges labelled at two or more separated regions	30 / 55 = 54.5%	**not attributable to single actions**

Primary-label counts: `Fuse` 17, `PrecisionChange` 15, `HostDeviceOverlap` 7, `Other` 5, `ChangeLaunchConfig` 3, `Vectorize` 3, `EliminateMaterialization` 2, `Tile` 1, `LoopTransform` 1, `IncreaseWorkPerThread` 1. Counting secondary labels as well, `WarpReduce` and `Vectorize` reach 6 occurrences each; `SharedMemoryTransform` never appears at all.

Two claims that do not survive this table.

1. *"No category ever produced the same diff twice."* All 55 diff fingerprints are distinct, so `distinct_diffs(label) == occurrences(label)` for **any** labelling, including a random one. The column cannot discriminate between hypotheses; the calibration record accompanying this paper carries the audit of this identity (`identity_audit`).
2. *"Frequent categories recur across tens of different tasks."* We constructed the null it requires — hold the label multiset fixed, permute labels across edges, recompute distinct-task counts (4,000 permutations). The observed values sit at chance: `Fuse` 16 distinct tasks against a null mean of 16.03 (66.8th percentile), `PrecisionChange` 13 against 14.26 (15.5th percentile). Cross-task recurrence is therefore **not** established by this sample, and the transferability argument in Section 2.5 loses its empirical support here. It remains a hypothesis that Section 4 would have to test with a design that does not cap edges per task.

What survives. Two things, and they are the load-bearing ones.

- **Compression relative to a faithful description.** Against the joint coarse descriptor of the same 55 diffs (hunk count × changed-line bucket, 24 cells), the label distribution costs 2.67 bits versus 4.26 — a factor of 1.6 — while using only 10 categories to name what the program did. Single low-cardinality summaries are cheaper still (hunk count alone: 2.49 bits) but they are not *descriptions* of the change; they are one projection of it, and Section 3.6 shows they carry no information beyond their own granularity.
- **Coverage without forced classification.** 90.9% of edges fell into the frozen vocabulary when `Other` was actively invited and each `Other` was required to explain why all eleven specific categories failed. A vocabulary that has to be inflated to absorb its data would have shown up as a long tail of `Other` with heterogeneous reasons; what we got instead was five identical refusals pointing at one missing category.

The claim worth keeping is qualitative, not count-based. Since the diversity *counts* are void, we instead examine realizations directly. All edges carrying one category, with the structural facts of each realization, are in the exemplar record accompanying this paper; an excerpt:

edge	task	lvl	× vs parent	hunks	changed lines	regions
`L2_2_11_167_168`	2_11	L2	1.06	3	141	3
`L2_2_48_29_30`	2_48	L2	1.57	1	107	1
`L2_2_28_106_107`	2_28	L2	2.60	4	55	3
`L2_2_42_1_69`	2_42	L2	26.83	2	120	2
`L2_2_60_112_114`	2_60	L2	1.13	7	81	4
`L3_3_16_13_15`	3_16	L3	1.56	7	149	3
`L3_3_2_33_61`	3_2	L3	7.69	4	189	3
`L3_3_43_2_19`	3_43	L3	1.84	1	60	1

One category, `Fuse`, spans 25 edges over 22 tasks, 23 distinct region sets, one-to-seven hunks, and 55-to-268 changed lines. Reading the rows, the named decision is the same — collapse two passes and drop a materialized intermediate — while its

textual realization shares almost nothing in size, shape or location. That is a much weaker statement than "the mapping is provably many-to-many", and it is stated as observation rather than measurement: anyone can open these eight diffs and disagree about whether one decision is being described.

Two structural facts also remain: `WarpReduce` and `SharedMemoryTransform` are unobservable in a Triton-only corpus and are exactly the two categories that never act as primary labels — evidence that granularity is pinned to the abstraction level of the emitted language. And **0 of 55 edges carry a single label**, which is why the 54.5% multi-region figure cannot distinguish "one action, several sites" from "several actions, one site each". The cross-site property that Section 2.2 makes central to the thesis is, on this data, *unidentifiable* — a design obligation for Section 4, not a finding.

3.6 Result: the informative test

Low entropy by itself is uninteresting, and a surface reading argues exactly that: coarse *surface* descriptions of the same diffs are equally low-entropy (hunk counts, 2.49 bits; changed-line buckets, 2.40 bits), so if the argument for the coordinate change were "transformations compress", it would already be refuted by counting hunks. What is needed is a test of whether the semantic label carries information about the *outcome* that surface structure does not.

Take the quartile of realized speedup versus parent as the outcome ($H = 2.00$ bits, 4 cells) and ask how much of it each descriptor explains. Raw mutual information is biased in favour of descriptors with more cells, so the comparison must be calibrated. Calibrating only one side gives the opposite answer.

Why the comparison must be calibrated symmetrically. The Miller–Madow bias term for a *marginal* entropy is $(k_X - 1)/2n$; the bias of a plug-in mutual-information estimate is $(k_X - 1)(k_Y - 1)/2n$. With $k_Y = 4$ the first is three times too small, and it under-corrects the 10-cell label descriptor far less than the 24-cell coarse descriptor. A comparison that applies the marginal term, or that calibrates only one of the two descriptors against its own null, therefore systematically favours whichever side was left uncorrected and can invert its own conclusion.

The test. We estimate mutual information three ways and calibrate both descriptors symmetrically, holding the descriptor fixed and permuting the outcome (20,000 permutations, seed 20261002).

Descriptor	cells	MI plug-in	MI Miller–Madow	null mean	null SD	permutation p
primary semantic label	10	0.697 bits	0.624 bits	0.397	0.072	**0.0005**
coarse diff structure	24	0.937 bits	0.792 bits	0.990	0.085	0.752
hunk count alone	9	0.367 bits	0.276 bits	0.361	0.071	0.436

This reverses the uncalibrated comparison, and it does not hinge on a choice of bias formula: the permutation test needs no correction at all, and under it the semantic label sits far above its own null ($p = 0.0005$), while the coarse descriptor's observed mutual information is *below* its own null mean (0.937 vs 0.990) — its apparently superior raw number is over-fitting to 24 cells, 11 of which hold a single edge. The paired

bootstrap difference $\mathrm{MI}_{\mathrm{label}} - \mathrm{MI}_{\mathrm{coarse}}$ is -0.361 bits with a 95% CI of $[-0.689, -0.034]$: on raw, uncalibrated levels coarse structure still wins. Which comparison is the informative one is exactly what the permutation calibration settles, and the calibrated reading is given above so the disagreement can be checked directly.

The scope of this result is narrower than its phrasing. A label that predicts the outcome beyond its own granularity is a real result about this corpus: the semantic description of what a successful edge did carries information about how much it gained that the shape of its diff does not. But the outcome here is an unattributed speedup ratio on 55 edges, 53 of which also changed autotuning configuration (Section 3.7b), measured from corpus metadata we did not re-time, on a sample we have already shown cannot estimate distributions. It is an argument for running Section 4, not a substitute for it.

How to read the table. The quantity that discriminates is the *excess over each descriptor's own null*, not the raw mutual-information level, because a plug-in estimate of mutual information rises with the number of occupied cells; both descriptors are computed on the same edges with the same outcomes, and the permutation calibration makes each null specific to its own granularity. Plug-in and Miller-Madow-corrected levels are both reported so the choice of correction need not be taken on trust.

3.7 Two findings that cut against the design

(a) `Other` is a single systematic hole, not noise. All five `Other` edges are the same decision: stop implementing the operation yourself and delegate it to a library backend — `x.matmul` / cuBLAS, `F.scaled_dot_product_attention`, `F.mish`, `F.linear`. They include the two largest realized speedups in the sample ($327\times$ and $69\times$ versus parent). The frozen vocabulary describes *how to restructure a program* and has no slot for *whether to implement it at all*.

Choosing a library call over a hand-written kernel is not an obscure insight, and it is not even absent from the strongest system in this area — but that system handles it the other way round. CUDA Agent's skill file *regulates* backend choice instead of offering it as an action: third-party libraries are forbidden except cuBLAS for GEMM and cuDNN for convolutions, cuDNN is made mandatory for Conv/ConvTranspose, and a context manager aborts execution on any fallback into `torch.nn.functional`, so that gains "originate solely from the generated CUDA kernels" [1, App. "SKILL.md"]. A decision worth 327x in our sample is therefore encoded as a constraint over admissible solutions rather than as something the agent reasons about, which is how it fell outside our vocabulary as well. The transferable part of the finding is narrower: **an action vocabulary assembled by introspection missed the highest-payoff decision in its own data, and the pre-registered numeric threshold (25% `Other`) failed to notice, because the refusals were homogeneous.** Any proposal to enumerate a transformation space should be tested against exactly this failure mode — by measuring what the refusals are *about*, not how many there are.

(b) Autotuning contaminates nearly everything. 53 of 55 edges co-modify `@triton.autotune` — adding it, removing it, rewriting the candidate list, or changing the cache key. In this corpus, structural transformation and hyperparameter search are

empirically inseparable, with two consequences. For attribution: a claimed speedup from `Fuse` is confounded with a simultaneously changed launch configuration, and our annotators repeatedly recorded exactly this inability to assign a primary cause (only 14 edges were labelled high-confidence; 38 medium, 3 low). For method: any action-level credit assignment scheme built on trajectories of this kind must first *factor autotuning out of the action space*, or it will learn a mixture of two different things. This is not a novel requirement — CuAsmRL separates autotuning from schedule search by construction [30] — but it is one that a vocabulary designed from source-diff inspection will silently violate, because `num_warps` and `num_stages` look like launch parameters and are written like code.

A related integrity note: 9 of 55 edges show more than 10x improvement over parent, and for at least two of those the parent was already at roughly reference speed (`fast_p` near 1.0). Enormous relative gains in an MCTS search corpus can be artefacts of a pathological parent or of timing methodology rather than evidence of optimization ability, and the timing pipeline is not audited here.

3.8 Verdict on the prerequisite

The prerequisite question is answered **yes for existence, no for the two properties we wanted to inherit from it**.

- **Existence (supported).** Successful schedule-level optimizations compress onto a small recurring typed vocabulary: 10 categories, 90.9% coverage against an actively invited `Other`, 2.67 bits versus 4.26 bits for a faithful coarse description of the same diffs. And the labels are not a decorative partition: they predict the realized speedup quartile beyond their own granularity ($p = 0.0005$) where hunk structure does not ($p = 0.752$, $p = 0.436$).
- **Cross-task transferability (not supported).** Distinct-task recurrence sits at chance under the label-permutation null (Section 3.5). The sampling design caps it, so no transferability claim is made.
- **Cross-site coordination (unidentifiable).** With 0 of 55 edges carrying a single label, the multi-region statistic cannot separate one distributed action from several local ones. The property that motivates the whole formalism is currently unmeasured, not confirmed.
- **Completeness of the vocabulary (falsified in one place).** One missing decision — backend choice — absorbed every refusal, and the numeric red line designed to catch this did not fire.
- **No reach into search or learning.** Nothing in Section 3 involves a policy choosing anything. The search and learning claims of Section 1.4 are untouched.

Section 3.6 alone does not establish the programme, and Section 3.5 alone does not refute it. What the data supports is that a transformation vocabulary is a *real, informative, incomplete* description of past successes — which is precisely the status that makes Section 4 worth specifying rather than optional.

3.9 Figures

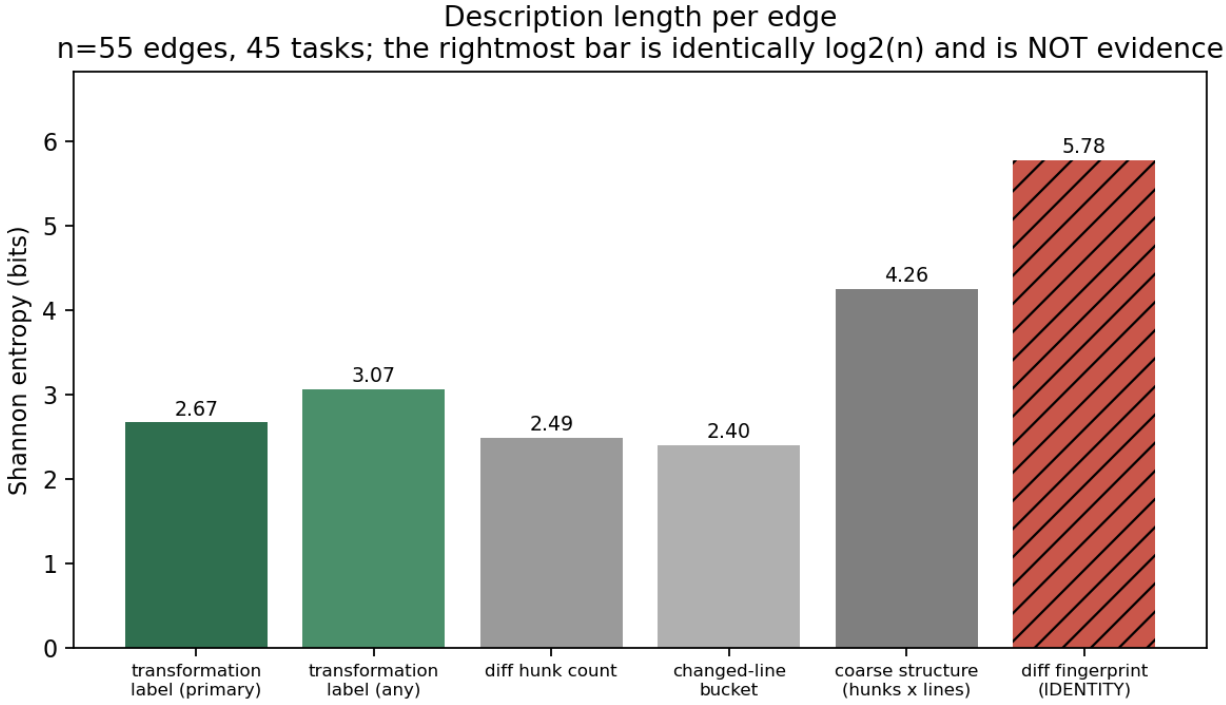


Figure 1: Description length per edge. The rightmost bar is identically $\log_2 n$ and carries no evidential content; the comparison that matters is against the 24-cell coarse structure.

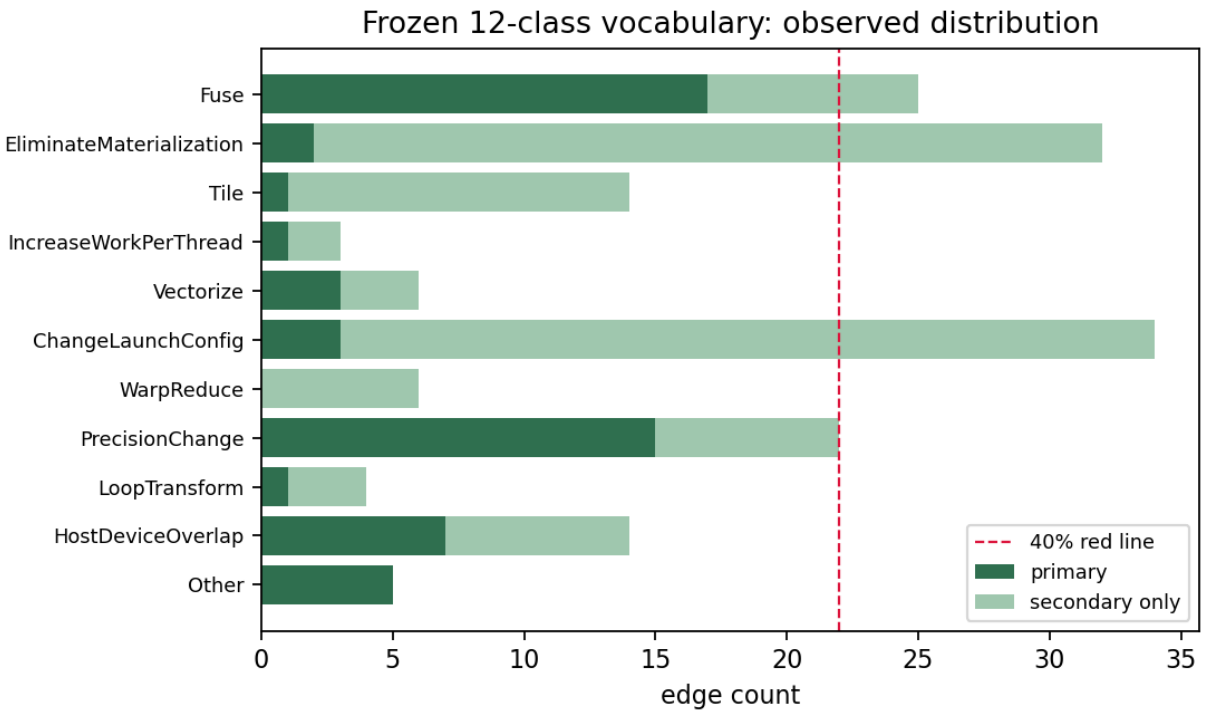


Figure 2: Observed distribution over the frozen 12-category vocabulary, primary versus secondary labels, with the 40% red line.

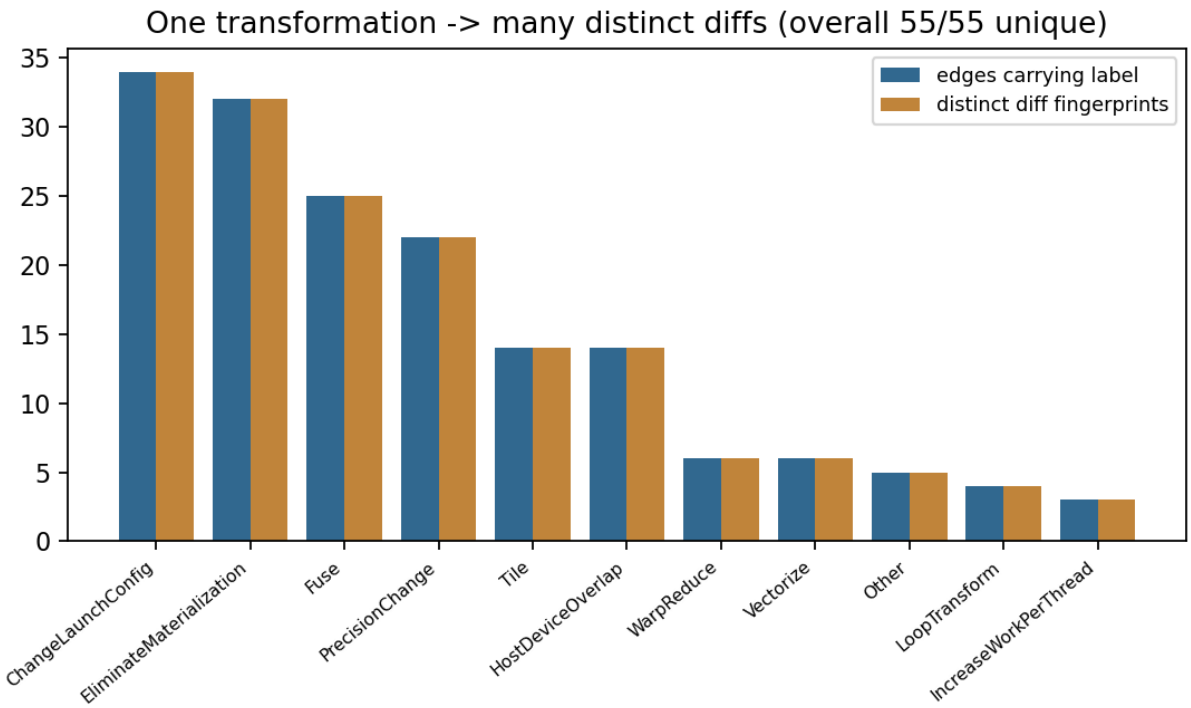


Figure 3: Occurrences versus distinct diff fingerprints per category. The two are equal by identity, which is why this panel cannot support the many-to-many claim.

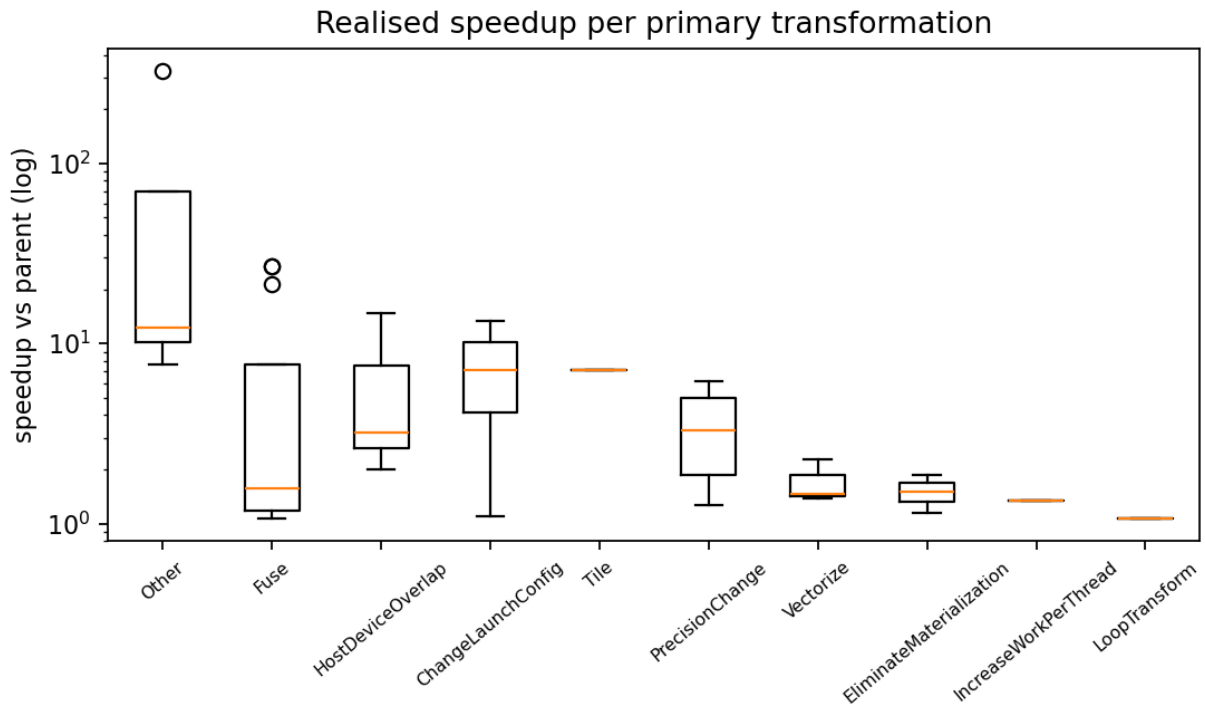


Figure 4: Realised speedup per primary transformation (log scale). The extreme values are dominated by pathological parents and by the backend-substitution category.

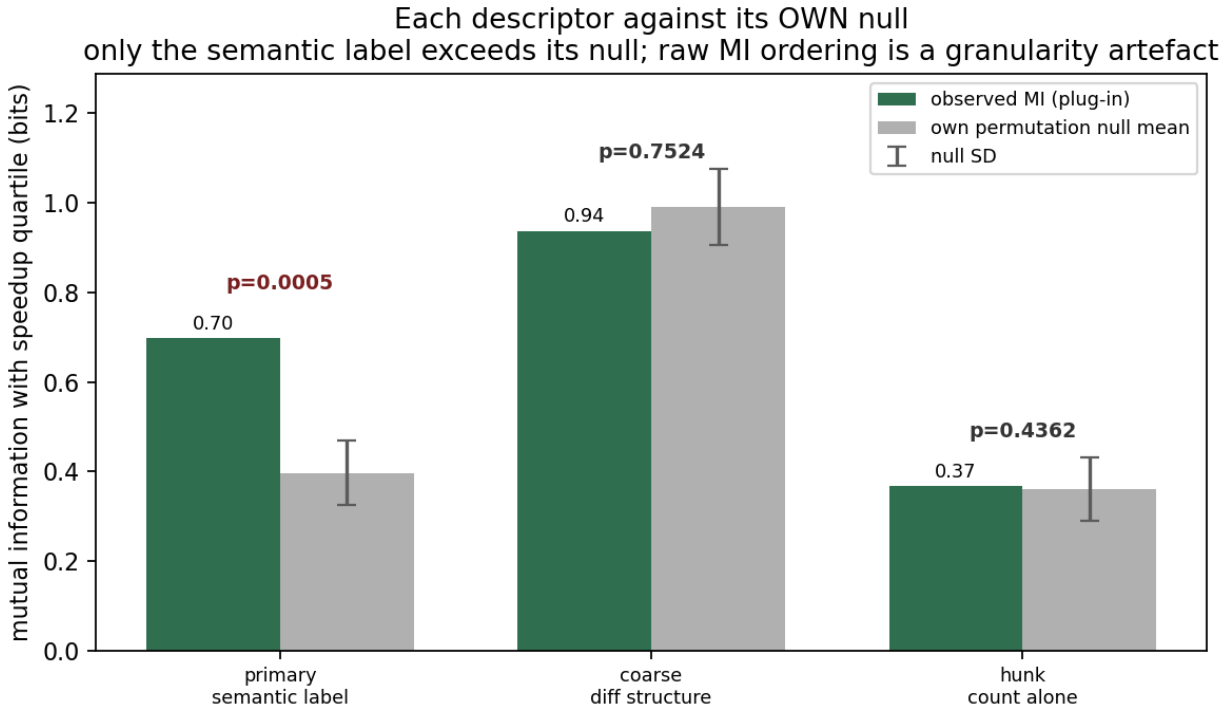


Figure 5: Each descriptor's mutual information with the realised-speedup quartile against its own permutation null. Only the semantic label exceeds its null.

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4. A Falsifiable Experimental Framework

4.1 The confound that must be defeated

The most tempting pseudo-result in this area is: *"let the model plan before writing code and accuracy improves"*. That establishes nothing about action spaces. It may be ordinary plan-first behaviour, or simply more reasoning tokens. Any experiment that cannot separate "planning helps" from "this particular planning representation helps" should not be run, because its positive outcome will be misread.

4.2 Five conditions

Condition	Interface	What it measures
Direct	generate the optimized whole kernel	baseline
Free-plan	write a free-text plan, then generate the kernel	the value of planning itself, and of any implicit abstraction the model already performs
Structured-action	select parameterized transformations from a frozen A, then realize them	the value of the *externalized structure*, net of planning

Condition	Interface	What it measures
Oracle-action	the correct transformation is supplied; the model only realizes it	whether strategy or realization is the binding constraint
Brute-search	enumerate a known transformation space and rank candidates with a learned cost model, no LLM in the loop	whether an LLM choosing transformations adds anything autotuning already had

Free-plan is the control that most published "decompose the task" claims omit. Oracle-action is the one that makes the framework self-terminating. Brute-search tests the strongest objection to the programme: that a well-engineered enumeration over a known schedule space with a learned cost model already finds what an LLM is asked to name (Section 5.3). If that enumeration suffices, moving the action unit is pointless. The other four conditions cannot settle it, since each keeps an LLM in the loop; without this arm the protocol can only ever compare variants of *LLM-in-the-loop* interfaces to each other.

Two prior systems occupy positions this table claims to distinguish, and the protocol must be read accordingly. K-Search already *"explicitly decouples high-level algorithmic planning from low-level program instantiation"* and reports an average $2.10\times$ over state-of-the-art evolutionary search on FlashInfer kernels, up to $14.3\times$ on complex MoE kernels [29] — that is a realized Structured-action condition with a positive result. CuAsmRL trains a reinforcement agent over an explicit action space of SASS schedule mutations, and separates autotuning from schedule search by construction [30]. So Section 4 is not a proposal for a system; it is an instrument for *independently falsifying* class results of that kind, by supplying the controls those systems do not run against each other.

The turn-level cut (Section 2.1) already exists in kernel RL, where multi-turn credit attribution is reported to matter [32]. Structured-action must beat *turn-level* decomposition, not merely token-level generation, or the abstraction has been paid for twice.

4.3 Reading tree

Pre-committed interpretation of each outcome pattern. Outcomes are measured on held-out tasks, with success rate and search cost (rollouts, tokens, wall-clock to first valid-and-faster kernel) both reported.

Observation	Conclusion	Consequence
Free-plan $\sim$ Structured-action $>$ Direct	planning helps; structure adds nothing	**no paper here**; drop the abstraction claim
Structured-action > Free-plan > Direct	the typed action space supplies extra inductive bias	core positive result; proceed to learning-level experiments
Structured-action $>$ Direct but $\sim$ turn-level decomposition	the gain is hierarchy, not typing	re-aim at credit assignment over agent turns [32] rather than at a vocabulary

Observation	Conclusion	Consequence
Oracle-action $\gg$ Structured-action	abstraction is useful, but *selection* is the bottleneck	reformulate as a policy-over-actions problem; $Q(P, a)$ becomes the object of study
Oracle-action $\sim$ Direct	telling the model what to do does not help	**direction is dead**: strategy/realization separation buys nothing, so no action space can help
Structured-action < Free-plan	the imposed granularity is actively harmful	investigate coupling: good optimizations may not decompose
Structured-action $\sim$ Brute-search	an LLM adds nothing over enumeration with a cost model	**the systems community already owns this problem**; re-target or withdraw
Oracle-action not runnable	φ cannot be constructed for this task family	**indeterminate — do not read as a null result** (see below)

Oracle-action and the realization obligation of Section 4.5 both require $\varphi : \Delta P \to a$, the map that recovers the intended transformation from a change, which this paper does not construct (Section 7.3). A protocol whose decisive branch depends on an object we have not built would report "direction is dead" whenever that object is unavailable, which is a false negative dressed as rigour. Oracle-action therefore carries an explicit precondition: a φ validated on a held-out set, or a human-supplied oracle standing in for it with the substitution recorded. If neither is available, the branch is *not evaluated*, and the paper's conclusion must say so.

The last two rows are the point of the protocol: it can lose, and a loss is informative.

4.4 Vocabulary freeze and held-out protocol

Build the vocabulary on one split of trajectories, **freeze it**, then test whether actions abstracted from past successes help unseen kernels. Feeding the same edges back as labels is not transfer, it is relabelling the answer.

This discipline guards against proxy substitution. When a difficult concept enters the question, the easiest move is to substitute a proxy that can be coded, measured and signed off, and then let engineering completeness hide the fact that the original question is still untouched: "how do we find the important optimization in the code" becomes "how do we apportion credit given an A/B result", and "understand program meaning" becomes a hand-written AST and regex classifier that quietly encodes the answer in its rules. A study that produces a benchmark, an index and a protocol but never reopens the original question has produced infrastructure and should be described as such.

The operational test for each experiment:

> **If this experiment completed perfectly, which judgement about the original research question would it change?**

If the answer is "the tooling is better" or "the records are more reproducible", the work is infrastructure and must not be presented as progress. Section 3 changes

the belief about vocabulary existence, and via 3.6 about whether entropy arguments work; an action-space schema and validator would change neither, which is why neither is built here.

4.5 Pre-registered confound controls

Drawing on Sections 3.7 and 3.6, any instantiation of Section 4 must add:

1. **Autotune isolation.** Autotune candidate sets fixed identically across all conditions, or explicitly modelled as a separate action dimension. Without this the comparison is between "structure" and "structure plus hyperparameter search", which is the wrong contrast; 53 of our 55 edges violate it by construction (Section 3.7b). CuAsmRL provides a working precedent by scheduling *after* autotuning rather than jointly [30]. Two ways this control fails in practice, both of which produce misleading results: pinning the candidate *list* does not pin the selected configuration, so the arms can still land on different block sizes; and shrinking the list to make it shareable can floor every arm at once, manufacturing a spurious null.
2. **Structure-matched baseline.** Include a condition that plans in an equal-length *structured-but-semantic-free* format (e.g. fixed slots filled with arbitrary content) so that "it is a template" and "it is a semantics" cannot be conflated - this is the control that Section 3.6 shows is necessary.
3. **Realization obligation.** Report, for each selected action, whether the subsequent edit actually touched the sites the action requires. An action that is chosen but not realized is a selection failure, not a coding failure, and the two must be counted separately.
4. **Held-out leakage audit.** Verify that no task used to build the vocabulary appears in evaluation, and that the vocabulary was not edited between the two.
5. **Decomposability stress test.** Sample edges where several categories co-occur (38 of our 55 are medium confidence for this reason) and check whether joint selection beats sequential selection. This is the direct test of the coupling objection in Section 7.

4.6 What the framework cannot be used to claim

Even a clean Structured-action win would establish the *search* claim only. The *learning* claim - that a policy over transformations is more sample efficient than a token policy - requires actual training comparison under matched model, matched trajectories, and matched terminal reward, with only the credit target changed. We do not have the infrastructure for that here, and no observational study can substitute for it.

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5. Implications Beyond CUDA

5.1 Edit-based and transformation-based code optimization are closer than a clean distinction suggests

The nearest neighbours are systems that learn to make performance-improving *code edits* [4,5,31], that treat the edit *format* as a training decision [6], and that learn program evolution from repository history [7]. The clean three-way distinction — typed semantics, cross-site coordination, cross-task vocabulary reuse — is cleaner than the evidence supports: two of those three legs do not survive the measurements of Section 3.5, so the defensible position is narrower.

What remains distinguishable:

- **Against edit-level work [4,5,31].** AEGIS decomposes slow→fast pairs into reusable *atomic edits* [31], and PIRE learns function-level performance edits [4]. Both are operation-level, and neither is *typed with parameters whose realization obligation is defined across sites*. What this paper adds to that comparison is a measurement rather than an idea: whether a fixed 12-category typed vocabulary covers real successful edges without being inflated. It does, at 90.9%, and the residual is homogeneous in a way that indicates a missing category (Section 3.7a).
- **Against transformation-level systems [29,30].** K-Search decouples algorithmic planning from program instantiation and gets large gains on kernels "requiring coordinated, multi-step structural transformations" [29]; CuAsmRL defines an explicit RL action space over SASS schedule mutations and even uses it to surface optimization moves automatically [30]. Neither, however, reports a *coverage measurement against a frozen vocabulary on public trajectories*, and neither reports the kind of null calibration of Section 3.6. Our residual claim is the descriptive one plus the protocol that could falsify the positive results.
- **Against format-level work [6].** AdaEdit shows the emitted representation changes latency and cost by over 30% at matched accuracy. That supports the general shape of "the coordinate system is not neutral", but it is a claim about *cost*; ours has to be about *inductive bias*, which is stronger and testable only by intervention.
- **Against turn-level decomposition [32,35].** Kevin attributes reward per agent turn for CUDA kernels and reports correctness 56%→82%; the wider credit-assignment literature increasingly works at segment and step granularity [35]. The field has already moved up from tokens one level. A transformation vocabulary must earn its keep over *that*, which is why Section 4.2 includes turn-level decomposition as a comparison point rather than treating token-level generation as the only baseline.
- **Against plan/action unification [33,34].** Two agent-side results make a representation claim here, and they cut against us. CodeAct replaces text/JSON tool calls with executable Python as a *unified action space* and reports up to 20% higher success across 17 models [33] — direct evidence that the action representation is not neutral, which is the shape of our thesis minus its ker-

nel specificity. ReCode goes further and dissolves the boundary between planning and acting altogether, treating high-level plans as placeholder functions recursively expanded into primitive actions with dynamic granularity control [34]; that is, in another domain, close to the Structured-action condition of Section 4. Neither is about GPU kernels, neither freezes a typed transformation vocabulary, neither reports whether such a vocabulary covers real successful optimizations, and — crucially — neither runs a Free-plan control of the kind Section 4.1 insists on. The consequence is that the Section 4.3 branch Free-plan $\approx$ Structured-action $\Rightarrow$ no paper is a live risk that adjacent fields have not resolved either, and what is left to claim is: the descriptive measurement in Section 3, the falsification discipline, and the kernel-specific confound controls that only this domain forces (autotune isolation, backend substitution).

- **Against a named transformation vocabulary that already exists in practice [1].** The strongest existing system already instructs its model in exactly our vocabulary, as prose. Its skill file orders optimizations by expected impact — Priority 1 algorithmic (>50%): kernel fusion to reduce memory traffic, shared-memory tiling for reuse, memory coalescing; Priority 2 hardware utilization (20-50%): vectorized `float2/float4` loads, warp-level `__shfl_sync/__ballot_sync` primitives, occupancy tuning by block size and registers; Priority 3 fine-tuning (<20%): instruction-level parallelism, FP16/TF32 mixed precision, prefetching and double buffering — and appends a checklist covering bank-conflict avoidance by padding, loop unrolling, WMMA/MMA tensor cores, persistent kernels and CUDA Graphs. It even prescribes the expectation-recording habit we would need for credit assignment: "Document expectation: 'Fusion will eliminate 3 kernels, expect ~20% speedup' " before implementing. This weakens the novelty claim and supports the premise. Somebody who ships the strongest published system in this domain already believes optimization behaviour is describable as a small, ranked, named set of program transformations. What that system does *not* do is make the vocabulary the interface of learning or search: the enumerated names live in a prompt, the policy is still trained over tokens, the tiers are advice rather than typed parameters with realization obligations, and no one has measured whether that checklist covers real successful optimizations or merely sounds right. Our Section 3 is that measurement, and Section 4 is the protocol for the part their design leaves untouched.

The difference can also be stated without relying on the vocabulary being correct: edit-, turn- and plan/action-unification methods take their unit from the *interface* (what a diff, a tool call, or a recursive function expansion happens to be), whereas a transformation takes its unit from the *program's performance model* (what the hardware will actually do differently). That is a clean distinction, and it is also the reason the distinction is dangerous — a performance model is exactly what an LLM is worst at being assumed to have.

5.2 Hierarchical RL has a name for this, and a warning

Temporal abstraction is old technology: options [11], the option-critic architecture [12], and skill libraries in LLM agents such as Voyager [14]. Our $a = (\tau, \ell, \theta)$ is an option with parameters. That lineage brings two things we should inherit rather than

re-derive: the formal machinery for policies over sub-policies, and a candid literature on *when higher-level actions fail to help*. Recent work on option-critic pathologies reports policy necrosis and redundant coverage - learned options that are never used or that overlap so much that the hierarchy buys nothing [13]. That literature is why Section 4.3 pre-commits negative and indeterminate branches rather than only a winning one. The present data does not decide between them: the semantic label is informative about realised speedup beyond surface structure (Section 3.6), and whether that survives independent labelling is open (Section 7.1). The hierarchy is therefore neither confirmed nor shown redundant — it remains exactly the open question that [13] and Section 4 are about.

5.3 The compiler lineage is a warning too

Halide separated schedule from computation [8]; TVM learned to search over schedule transformations [9]; Ansor searches over tensor-computation factorizations [10]. These systems already treat program transformation as the unit of optimization work. Two lessons follow.

First, the idea is not new, and any novelty claim must rest on the LLM interface, not on the existence of a transformation space. Second, and more importantly, there is an unresolved tension between our thesis and the compiler's practice. Autotuning systems deliberately *do not* ask the programmer to name the transformation: they enumerate a space and search it with a learned cost model. If that already finds the schedule, an LLM choosing transformations must be adding something the cost model lacked — handling the parts of the space no one wrote an enumeration for. That is a real hypothesis with a real chance of being false, and it is the strongest objection we know of to this direction.

None of the first four conditions can answer that objection: all four keep an LLM in the loop, so the protocol could only ever rank variants of itself. Section 4.2 answers it structurally, with a fifth arm, **Brute-search** (enumeration plus learned cost model, no LLM), and Section 4.3 adds its reading row: if Structured-action is statistically indistinguishable from Brute-search, the systems community already owns this problem and this programme should be re-targeted or withdrawn. Note that this is a test we expect to be *hard to win*: a competent Ansor-style search over a well-specified schedule space is a formidable baseline [10], and an LLM earns nothing here by being more expressive.

There is a deeper asymmetry that the compiler analogy conceals, and it is the thesis's weakest joint. Between assembly, IR and a high-level language sit compilers: relations that are effectively total functions, so an abstraction layer can be *deleted* and re-derived with confidence. Sul and Ré's own version of the retirement argument depends on exactly this, and they name the deal honestly — "we took that deal because the compiler was right more often than we were" [36]. Between a named transformation and its code, however, sits an LLM, and the relation $a \to \Delta P$ is a stochastic map, not a function. Everything in this paper that treats a transformation as an object — including the credit assignment that motivated it — inherits that gap. Two consequences follow: realizing a chosen action is a *first-class failure mode* rather than an implementation detail (which is why Section 4.5 control 3 requires reporting it sepa-

rately), and the abstract's $\varphi : \Delta P \to a$ in the reverse direction is no more guaranteed to exist than the forward one. If realizers are unreliable, a transformation-level action space buys nothing no matter how well the vocabulary fits, and that is exactly the outcome Section 4.3's Oracle-action branch is designed to catch.

5.4 Agent skill design is the same question in a different costume

A related debate concerns the long natural-language instruction files (skills) attached to coding agents. Systems exist that force a model to emit a structured intermediate representation — a hierarchical graph with declared persistent state and annotated data edges — before writing code, then have a deterministic auditor check it [28], and ship very large hand-maintained instruction files alongside the method.

Two observations from Section 3 transpose to that design, and neither has been measured. First, quantitative requirements dominate qualitative ones in practice: "produce ten diagrams" is easy to satisfy, "each diagram carries information" is not — so an agent asked to *plan* will produce plan-shaped text rather than decision-shaped text, which is exactly the failure the Free-plan control exists to catch. Second, compliance appears to degrade as an instruction file grows, until no single rule is reliably followed. If the structured-action condition were implemented as a long prose catalogue of transformation types, it would inherit both problems and Free-plan could beat it for attention-budget reasons rather than semantic ones. That argues for the small typed interface of Section 2.3 rather than documentation — and Section 4.5 control 2 is what would test the second claim instead of assuming it.

5.5 Verification is a prerequisite, not an appendix

Every experiment in Section 4 requires a trustworthy correctness-and-timing oracle, and this is not a solved problem in GPU kernel benchmarks. Reported failures include kernels that are numerically wrong yet pass the checker [20], benchmark oracles that are themselves weak under mutation testing [24], and results that do not survive verification against real workloads or against PyTorch under stricter measurement [22,23,25]. Adversarial hardening work closes the loop from the other side. The hacker-fixer loop alternates three agents — a hacker that tries to pass the verifier without solving the task, a fixer that patches the verifier to reject each discovered exploit, and a solver that confirms the patched verifier still admits legitimate solutions. Auditing 1,968 tasks across five terminal-agent benchmarks, it finds 323 (16%) hackable by frontier models given only the task description; on KernelBench it drives attack success from 62% to 0% on a held-out corpus of publicly reported exploits, and a *weaker* model running the loop defends against substantially stronger hackers (76% and 61% attack rates reduced to 0%) [19].

It follows directly from that setup that if candidate selection scores proposals by raw measured speedup, a spurious 100x exploit numerically dwarfs a legitimate 1.5x improvement, so the search is biased toward the exploit *before* any verifier is consulted. Bounding or categorizing the reward is therefore not benchmark hygiene but a structural precondition for transformation-level credit assignment — otherwise a value function indexed on actions will learn to prefer transformations that are structurally

unsound but numerically loud, and the abstraction will have imported the vulnerability rather than fixed it. The quantised reward of [1] already addresses part of this from the other direction — it is designed so that tasks with large headroom do not dominate selection — which is a reason to treat reward shaping as settled infrastructure and the choice of action unit as the open question.

5.6 The training-substrate criticism

Finally, the framing here suggests a different reading of the instability literature around long-horizon agentic RL. The published record contains a tension worth naming. On one hand, CUDA Agent states that despite strong general programming performance, LLMs remain uncompetitive with `torch.compile` for kernel generation, and motivates large-scale RL to close that gap [1]. On the other hand, most tokens in a CUDA kernel are ordinary programming tokens — `for`, `if`, `float`, `int`, `*`, `[]`, `+` — with only a small, highly patterned minority that is CUDA-specific, so a corpus-frequency account of why token-level training should be *unusually* unstable in this domain is hard to sustain on first principles. And the account is not domain-specific in the way that would make it a CUDA result: whatever rarity a target domain has in pretraining — an obscure programming language, a hardware description language, a formal proof language — should produce the same low-probability tail. If the explanation transfers to every rare domain, it cannot be the explanation of anything about GPUs. On that reading, low-probability-tail blowups are more plausibly a train–inference mismatch and out-of-distribution-context problem of long multi-turn tool use — swallowing compile errors, profiler dumps, and odd intermediate program states — than a statement about learning theory for CUDA syntax, which is consistent with the general finding that training-inference mismatch, not domain rarity, is what destabilizes LLM reinforcement training [26,27].

The specific base-model accuracy figures that circulate around this debate are not asserted here: the primary source does not report them, and its training-time observables are not released [1]. The argument above is therefore a *framing* claim — that the token-level interface is a plausible common cause of both the instability and the credit problem, because it forces one policy to model program structure and tool noise in the same sequence — not an empirical one.

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6. Related Work

Positioning against each cluster is argued in Sections 1.1 and 5.1-5.6; this section records only which work belongs to which cluster and what we take from it.

Agentic RL and CUDA generation. KernelBench [2]; CUDA Agent, the system this framing is aimed at: it trains a token-level PPO policy inside a tool-defined environment whose action space is Edit/MultiEdit-class operations, and it ships a priority-ranked named optimization checklist in its skill file [1]; CUDA-L1 (contrastive ranking over execution feedback) [16]; Dr. Kernel (reward design for Triton) [17]; CudaPerf (structural rewards from coalescing, occupancy, arithmetic intensity and synchronization, still at token level) [15]; NKI-Agent (the CUDA-Agent recipe on AWS Trainium,

with a negative GRPO result) [18]; Kevin (multi-turn RL, turn-level reward attribution) [32].

Credit assignment below the trajectory. A 69-paper survey corpus organised by credit granularity — trajectory, segment, step, token — with an identification analysis showing that text-only histories can leave even the *sign* of credit unidentified [35]. This is the cluster our Section 1.2 must be read against: the field reweights credit at finer granularity while keeping the action unit at the token.

Action representation in LLM agents. CodeAct consolidates an agent's actions into executable Python as a unified action space and reports up to 20% higher success rates [33]; ReCode unifies planning and acting recursively, with plans as placeholder functions expanded to primitive actions under dynamic granularity control [34]; Voyager persists acquired skills as reusable code [14]. These are the strongest general evidence that the coordinate system is not neutral, and the closest generic analogues to our structured-action condition (Section 5.1).

Transformation- and edit-level search over programs. The cluster that constrains our novelty claim most: K-Search, which explicitly decouples algorithmic planning from program instantiation and reports $2.10\times$ average, up to $14.3\times$ over evolutionary baselines on FlashInfer kernels [29]; CuAsmRL, an RL agent over an explicit SASS schedule-mutation action space, which also surfaces optimization moves automatically [30]; RAS/AEGIS, beam search with retrieved slow→fast examples and reusable *atomic edits* [31]; performance-improving code edits curated from 77k C++ submission pairs and evaluated in a full-system simulator to control timing noise [4]; neural models of edit sequences [5]; RL on software evolution [7]; and the finding that the emitted edit format is not cost-neutral [6]. Section 5.1 states the residual difference.

Compiler search spaces. Halide separates schedule from computation [8]; TVM learns over schedule transformations [9]; Ansor searches tensor factorizations [10]. Section 5.3 explains why this lineage is both our ancestry and our strongest objection.

Hierarchical RL and skill discovery. Options [11], option-critic [12], reported option pathologies — policy necrosis and redundant coverage [13] — and LLM skill libraries [14]. Our $a = (\tau, \ell, \theta)$ is a parameterized option (Section 5.2).

Benchmark integrity. Correctness illusion [20], mutation analysis of kernel-benchmark oracles [24], adversarial hacker-fixer verifier hardening [19], contract-grade verification [21], and stricter verification against real workloads or against PyTorch [22,23,25]. These bound any Section-4 experiment more than any other part of this paper (Section 5.5).

Agent harnesses and skills. Structured intermediate representations emitted before code and audited deterministically, shipped with very large instruction files [28] (Section 5.4).

Trajectory data. AdaExplore, a failure-driven diversity-preserving search method that releases its parent-child trees [3]. Used here as data about optimization behaviour; no search baseline of any kind was run in this paper.

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7. Limitations and Open Questions

7.1 The circularity of machine annotation, and what would actually control it

A human reading of thirty to fifty real edges is the control this question needs, for a specific reason: whether the vocabulary exists is the question, and the entity that invents the vocabulary should not also be the entity that decides whether the data fits it. Annotation here is automated instead. The consequence is unavoidable: the labelling procedure and the category system come from the same model family, so a `Fuse` judgement may be a statement about the annotator's habits of description rather than about the code.

Declaring the risk is not a control. The following criteria would be:

1. **Independent blind human annotation** of the same 55 edges by at least two annotators who did not author the vocabulary, reporting Cohen's κ or Krippendorff's α. With κ unreported, "10 categories cover 90.9%" and "the annotator could name almost anything" are indistinguishable.
2. **Bottom-up vocabulary construction from a disjoint source** — induce categories open-coded from a held-out half of the corpus by a different family (human or model), freeze, then test whether the pre-registered vocabulary is recovered. Convergent validity, not self-consistency.
3. **Survival of Section 3.6 on consensus labels.** The permutation result is the paper's main quantitative claim about the labels; if it vanishes when labels come from human agreement rather than from a single model family, the claim belongs to the annotator, not to the programs. This is the decisive test; what has been run so far uses agreement between two automated passes rather than human agreement (below).
4. **Label availability without vocabulary availability.** Give the annotators only the twelve names, not our definitions, examples, or parameter slots, to reduce the chance that category *semantics* are shared by construction.

Criteria 2, 3 and 4 have been executed. The measurements are reported below; criterion 1 (a genuinely independent, preferably human second annotation) has not, and remains the paper's main open requirement.

Definition-free relabelling (criterion 4). A separate annotator labelled the same 55 edges seeing only the twelve bare category names — no definitions, no parameter slots, no examples, no decision rules. Agreement with the original pass is substantial: 74.5% raw agreement, Cohen's $\kappa = 0.687$. Three categories went entirely unused by the blind pass (`WarpReduce`, `SharedMemoryTransform`, `HostDeviceOverlap`), which tracks the Triton-abstraction reading of Section 3.5 better than it tracks a labelling failure.

Survival under blind labels (criterion 3). The result does not stand alone. Recomputing the Section 3.6 permutation test on the blind labels gives MI 0.451 bits with $p = 0.106$ — not significant — where the informed labels gave 0.697 bits with $p = 0.0005$. Restricting to the 38 edges where the two passes agree restores significance (MI 0.664, $p = 0.0016$). The informativeness of the vocabulary is therefore partly a property of the elicitation: supplying the definitions sharpens labels into something that predicts outcomes, and we cannot yet separate how much of that is

structure in the programs from how much is guidance we provided. Criterion 1 — genuinely independent, preferably human annotation — remains unexecuted and is the only test that can separate them.

Bottom-up open coding (criterion 2, partial). A coder with no access to the frozen list induced 7 categories from 24 successful edges drawn from tasks disjoint from the annotated sample (zero task overlap; the open-coding record accompanying this paper). The recovered space is recognisably the same territory — work/layout remapping, precision regime, setup hoisting and caching, pipeline-stage reshaping, backend reassignment, algebraic reformulation, redundant data-movement removal — with two informative divergences. The coder made launch-constant adjustment strictly secondary (19 co-occurrences, 0 primary), suggesting our `ChangeLaunchConfig` over-partitions what the data treats as accompaniment; and the coder independently rediscovered backend reassignment, the same category we were forced into `Other`. That is convergent evidence about the *space* and about one hole in our list, not convergent validity for the twelve names as drawn, since coder and vocabulary still share a model family.

Two measures partly constrain the annotator: `Other` was explicitly invited and each `Other` had to justify rejecting all eleven specific categories, which is how the backend-choice hole surfaced; and the confidence distribution is reported — 14 high, 38 medium, 3 low, so the majority of these labels are ones the annotator was not sure about.

7.2 The three dangers that would close the programme

Each of the following would kill the direction, and each is sharpened by the measurements of Section 3.

(1) Good optimizations may not be discretely decomposable. A strong kernel rewrite is often a *whole-design* change: tile size depends on layout, layout on thread mapping, thread mapping jointly with shared-memory footprint and occupancy. Forcing that into an action *sequence* may destroy exactly the joint search that produced the win, and post-hoc decomposition into `Tile + Fuse + Vectorize` may be interpretation rather than explanation. Evidence in this corpus: multi-label edges are the norm, and the most frequent annotation note was inability to attribute the speedup to one action. Open question: does joint selection over a vector of parameters outbeat sequential selection, and if not, is the decomposition the right level at all? (Section 4.5 control 5.)

(2) The model may already do this abstraction privately. If a large language model internally represents "use a warp reduction" without being told to, then externalizing the action may only *print* reasoning that was already happening, reducing nothing. This is precisely why the Free-plan control exists, and why a result of the form Free-plan $\sim$ Structured-action $>$ Direct must be reported as a null. We have no way to settle it observationally.

(3) Selection may be easy and realization may be the bottleneck. Telling a model "use warp-level reduction" may still leave it unable to write a correct, fast implementation. If that is the common case, high-level actions are not the binding

constraint and the whole programme mislocates the difficulty. The Oracle-action condition exists to expose exactly this, and its neutral outcome is the pre-registered end of this research direction.

7.3 Sampling, attribution, and measurement limits

The annotated 55 edges are a multi-hunk-enriched biased sample and cannot estimate population label frequencies. **The eligibility gate itself is inherited from the corpus: every edge is conditioned on the benchmark's own `correctness == true`, and two of the papers we cite argue that this oracle admits false passes [20,24]. Sections 3.4-3.6 therefore rest on a filter we have shown reason to distrust, and a mutation-hardened oracle could change which edges count as successes at all.** Related measurement gaps: the ">5% faster" threshold has no noise floor because timing was taken from corpus metadata and never re-measured; `fast_p`'s baseline convention is not audited; 9 of 55 edges show $> 10\times$ gains over parent, at least two of which had parents already near reference speed, so some realized-speedup values used as the Section 3.6 outcome are plausibly timing artefacts; 53 of 55 edges change autotuning alongside structure, so per-action attribution in Sections 3.5-3.7 is confounded by construction. The trajectory corpus was produced by a *search* procedure rather than by a policy learning from a reward, so the visited-program distribution is not the on-policy distribution a Section-4 experiment would face. Finally, φ is never constructed here — the claim is that these diffs have labels, not that a mechanical map from diff to label exists — and the gap between those two is a separate research question (automatic action recognition), left open here.

7.4 Language dependence

Section 3.5's cleanest negative result concerns the vocabulary itself: two categories are unobservable because Triton hides the level of abstraction at which they live. A transformation vocabulary is therefore not a language-neutral object. It is pinned to the abstraction level of whatever the model is asked to emit, and "CUDA optimization" spans at least three levels (CUDA C++, Triton, high-level framework graph) with different natural vocabularies at each. Any transfer claim must be intra-level unless a mapping between levels is specified.

7.5 What this paper does not show

No search experiments. No training. No new model. No measured improvement in any metric. Section 3 is a calibrated existence-and-non-sufficiency result about description; Section 4 is a protocol that has not been run. The title is a question, and Section 3.6 answers it only provisionally.

---

8. Conclusion

We asked whether the token-level action space used by agentic reinforcement-learning systems for GPU kernel optimization is the wrong coordinate system, and whether parameterized program transformations $a = (\tau, \ell, \theta)$ are a more natural unit for both search and learning. The answer is: the premise holds, two of the arguments we built for it do not, and the question remains experimentally open.

The premise holds. Public successful optimization trajectories compress onto a small recurring typed vocabulary. On 55 annotated MCTS edges spanning 45 tasks, 10 of 12 frozen categories cover 90.9% of edges against an actively invited `Other`, at 2.67 bits versus 4.26 bits for a faithful coarse description of the same diffs. More importantly, the labels are not a decorative partition: under symmetric permutation calibration the semantic label is significantly informative about realised speedup (plug-in mutual information 0.697 bits, 0.624 after bias correction, $p = 0.0005$) while coarse diff structure ($p = 0.752$) and hunk count alone ($p = 0.436$) do not beat their own nulls. It is bounded by a control: relabelling the same edges without being given the category definitions leaves the same test insignificant ($p = 0.106$, against $p = 0.0005$; the two passes agree on 74.5% of edges at $\kappa = 0.687$), so part of the effect is elicited by the vocabulary rather than found in the programs (Section 7.1).

Two supporting arguments fail. The first was compression itself, presented in opposition to a diff-fingerprint entropy of 5.78 bits: with 55 pairwise-distinct fingerprints that quantity is identically $\log_2 55$, and the claim that "no category ever produced the same diff twice" is true of any labelling whatsoever. Both fail. The second was cross-task recurrence, which sits at chance under a label-permutation null (16 observed distinct tasks for `Fuse` against a null mean of 16.03) because our sampling caps edges per task. The transferability argument in Section 2.5 is therefore unsupported by this data, and the cross-site coordination property that motivates the whole formalism is *unidentifiable* here, since no edge carries a single label.

These two limits are independent of each other; the calibration behind the main result is recorded in Appendix B.

The question remains open, and now has a sharper shape. Observational evidence cannot decide a claim about interventions, and the interface-level question has already been partly answered by systems that got there before us: K-Search decouples planning from instantiation and reports $2.10\times$–$14.3\times$ gains [29]; CuAsmRL trains directly over an explicit schedule-mutation action space [30]; Kevin attributes reward per agent turn [32]. Section 4 therefore specifies a five-condition protocol — Direct, Free-plan, Structured-action, Oracle-action, and a Brute-search arm with a learned cost model and no LLM in the loop — with a reading tree whose neutral, inverted, and *indeterminate* branches are fixed in advance, including the branch that withdraws the direction if an LLM choosing transformations cannot beat enumeration, and the precondition that protects a false negative from being reported as a null when φ cannot be constructed.

Two further findings should travel beyond this specific topic. First, every edge we could not classify was the same uncategorized decision: stop implementing this yourself and call a library. An action space assembled by introspection missed the highest-

payoff action in its own data, and the pre-registered numeric threshold designed to detect exactly this failed to fire because the refusals were homogeneous — a warning that coverage percentages are the wrong instrument for detecting a missing category. Second, autotuning moved together with program structure in 53 of 55 successful edges, so in realistic kernel-optimization corpora hyperparameter search and structural transformation are not separable without being designed apart; anyone proposing action-level credit assignment for code should assume that mixture until shown otherwise.

The narrow contribution is thus: one calibrated measurement of a premise, an explicit retraction of the two supporting statistics we originally relied on, and a protocol capable of deciding what the measurement cannot. The broader claim we are willing to defend applies well past CUDA: before arguing that an abstraction is a better coordinate system, establish that it beats a null of its own granularity rather than a ceiling it cannot reach, check which decision your vocabulary silently omits, and do not trust a verification harness that has never been asked to question a method.

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Appendix A — Reproducibility

All Section 3 numbers are produced by three scripts in `qwen-transform-action-space/scripts/`, in order:

```
python scripts/select_edges.py            # 2,510 candidates -> 55 sampled edges + diffs
python scripts/analyze_entropy.py        # annotations -> entropy, reuse, red lines, figures
python scripts/calibrate_informativeness.py  # cardinality-matched permutation calibration
```

Random seed 20261002; no GPU required; input is the public AdaExplore archive [3]. Vocabulary and annotation protocol are frozen in `VOCABULARY_FROZEN.md` and `ANNOTATION_GUIDE.md`. Per-edge unified diffs are in `data/edges/`, annotations in `data/annotations.json`, calibration in `data/informativeness_calibration.json`. No classifier, locator, or AST pipeline was written for this study, by design; the only automated statistics are diff-structural (hunk count, changed-line count, region attribution).

AI usage disclosure. Literature retrieval, edge annotation, analysis, figure generation, and this draft were produced by an automated agent pipeline without human

verification of the manuscript content. Section 7.1 states the consequences for the central empirical claim.

Appendix B — Production record

This appendix records how the analysis behind Section 3 came to be stated differently at earlier stages, and what the automated review stage of this run could and could not establish. It is placed here rather than in the body because it bears on how much weight to give the numbers, not on the argument about action spaces.

These are not confessions of style; all three changed what we believe about the results above, and all three bear directly on the methodological claim this area is making about automated research.

(1) A verification harness can be numerically exhaustive and methodologically blind. Our integrity stage ran 69 assertions re-derived from raw data, and passed 69/69 while the headline statistical conclusion of Section 3.6 was wrong. It compared *paper number* to *script output* and never *script method* to *valid method*; a misapplied bias correction propagates through it perfectly, because the number it checks is the number the bug produced. The same harness also faithfully certified a figure that Section 3.5 now identifies as an identity. This is the failure mode Section 4.4 warns about — internal rigour increasing while the question stays untouched — reproduced inside our own quality gate rather than in someone else's infrastructure. The practical consequence for any pipeline that reports to itself: reconciliation checks must be paired with at least one check that can question a method, which in our case was the adversarial review stage, and which no self-review we could have scheduled would have substituted for.

(2) Our own retrieval failure was recorded as somebody else's fabrication. One reviewer quoted CUDA Agent's source — "these tools define the action space of the agent" — in support of a stronger version of our critique. We could not reproduce it: the preprint has one version and no HTML rendering, so we classified the quotation as fabricated, dropped it, and weakened Section 1.1 accordingly. The quotation was accurate. When we later fetched the arXiv e-print as a LaTeX source bundle and read it, the sentence was there verbatim, along with the tool list and the paper's own statement that its PPO action a_t is the token.

The failure was not credulity but a category error in the other direction: we treated "I cannot retrieve this format" as "this cannot be verified", and then applied a conservative-looking rule — discard the unverifiable — which happened to discard a *true and inconvenient-to-us* claim. Two consequences for any pipeline that cites at all: a rejected quotation should be re-attempted through a different channel (source archive, alternate host) before it is deleted, because a format limitation is not an evidentiary one; and the direction of the error matters — a policy that only ever trims claims *favouring* the paper will systematically under-report, and here it deleted accurate material. Our §1.1 is now written from the source rather than from the abstract, and the correction is itself part of what we want on the record.

(3) A five-seat panel from one model family is one reviewer, not five. Sul and

Ré's blog carries an aside that reads as a prediction about this paper's own quality gate: a multi-agent "three-way review" is a sensible architecture, but nothing guarantees the three reviewers are checking different things, and they may converge on the same blind spot [36]. Our Stage 3 panel has exactly that shape — five distinct mandates, one cognitive family — and we observed both the benefit and the failure. Two reviewers independently found the statistical error that four earlier stages of self-checking had certified, which is corroboration a single reviewer could not have supplied, and that alone justified the panel. But panel diversity of *persona* is not diversity of *origin*: on the evidence of incident (2), a fabricated citation from one seat is a fabricated citation from all five, and no amount of role labeling inside a single model family substitutes for an outside reviewer. We keep the five-mandate structure because it found real problems, and we refuse to describe it as independent review.